COVID-19 PANDEMIC

Time Series Analysis & Forecasting

"A time series model developed to forecast COVID-19 deaths and identify regions of rapid pandemic spread across the United States."

Tech Stack Python • Pandas • NumPy • Matplotlib • Seaborn • Statsmodels

Key Concepts EDA • Time Series Analysis • Rolling Statistics • Stationarity Testing • ARIMA Forecasting

AUTHOR: LEAH T. MOGOTSI

1.Project Overview

The COVID-19 pandemic caused unprecedented loss of life and economic disruption globally. The main objective of this project was to perform exploratory data analysis on real-world COVID-19 data, identify regions and time periods of rapid spread, and develop a time series forecasting model using ARIMA to predict future daily deaths. The dataset covers 3,342 US counties across 58 states and territories from January 22, 2020 to March 9, 2023.

The project followed a structured data science workflow, from raw data loading and cleaning, through exploratory visualisation, to time series modelling using ARIMA. The goal was not just to describe what happened, but to understand the pattern of spread, identify the hardest-hit regions, and build a model capable of forecasting future deaths.

2. Research Questions

2.1 Research Questions

In order to better understand the progression of COVID-19 across the United States, several research questions were formulated and explored through statistical analysis and visualisations.

1. How did COVID-19 cases and deaths progress over time at a national level?
2. Which US states were most severely affected?
3. During which time periods did cases and deaths accelerate the fastest?
4. Can a statistical model forecast future daily deaths?

3. Findings & Analysis

3.1 National COVID-19 Trends

The first step was to understand the overall shape of the pandemic at a national level. By summing all county-level data into daily national totals, we produced cumulative trend lines for both confirmed cases and deaths.

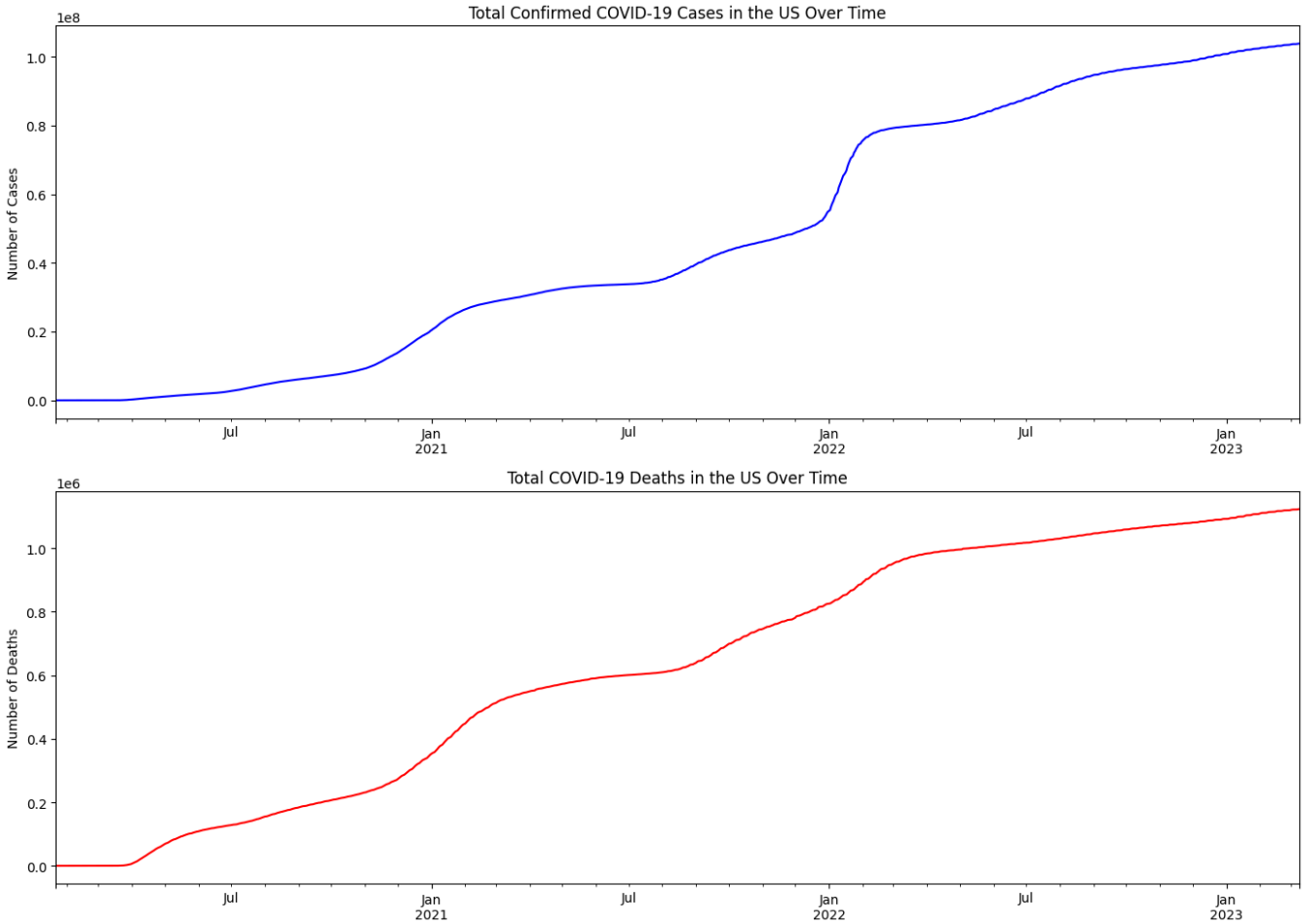

Figure 1: Total Confirmed Cases (blue) and Deaths (red) — US National, Jan 2020 – Mar 2023

Interpretation: Both confirmed cases and deaths show a clear upward trend from 2020 to 2023 with distinct waves. The most notable spike in cases occurred around January 2022. However, deaths did not rise as sharply during this period, suggesting vaccines were effective in reducing fatalities. By early 2023 both curves began to flatten, indicating the pandemic was winding down.

3.2 Regional Analysis — Top 10 States by Total Deaths

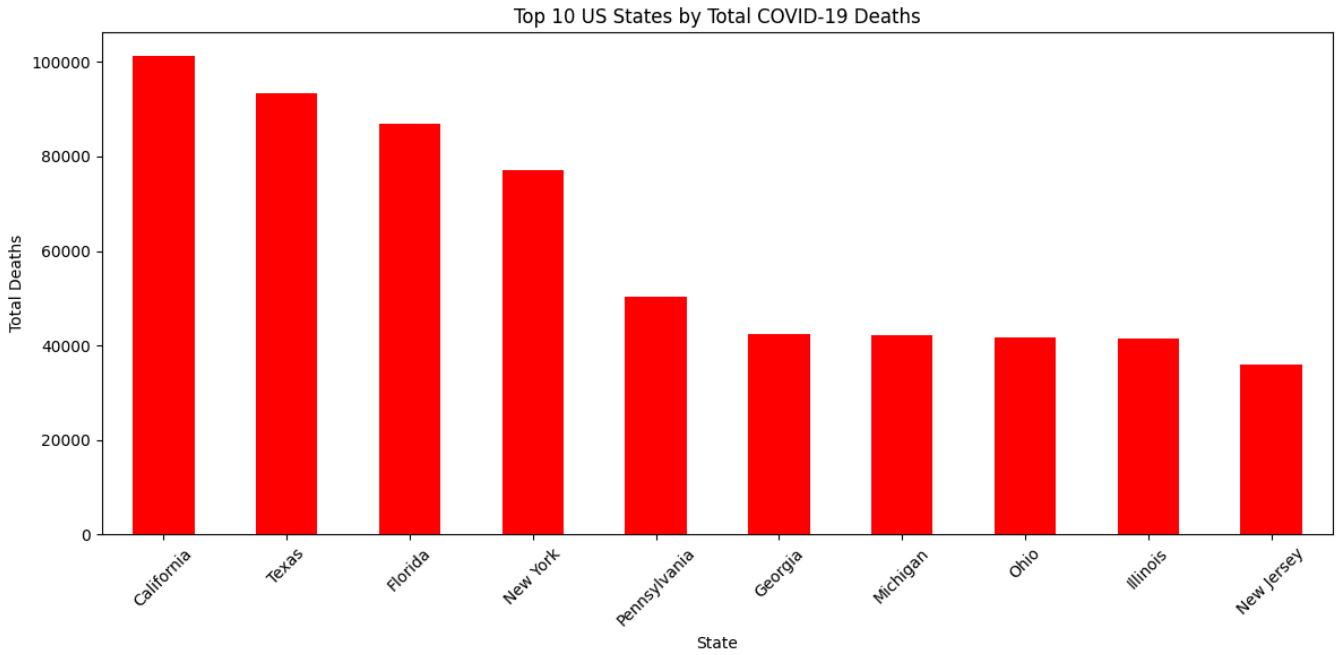

Figure 2: Top 10 US States by Total COVID-19 Deaths

Interpretation: California recorded the highest total deaths, followed by Texas and Florida. All top 10 states are among the most populous in the US, confirming that population density was a key driver of pandemic severity. This highlights the importance of targeted interventions in high density regions during future outbreaks.

3.3 Daily Death Patterns — New York Case Study

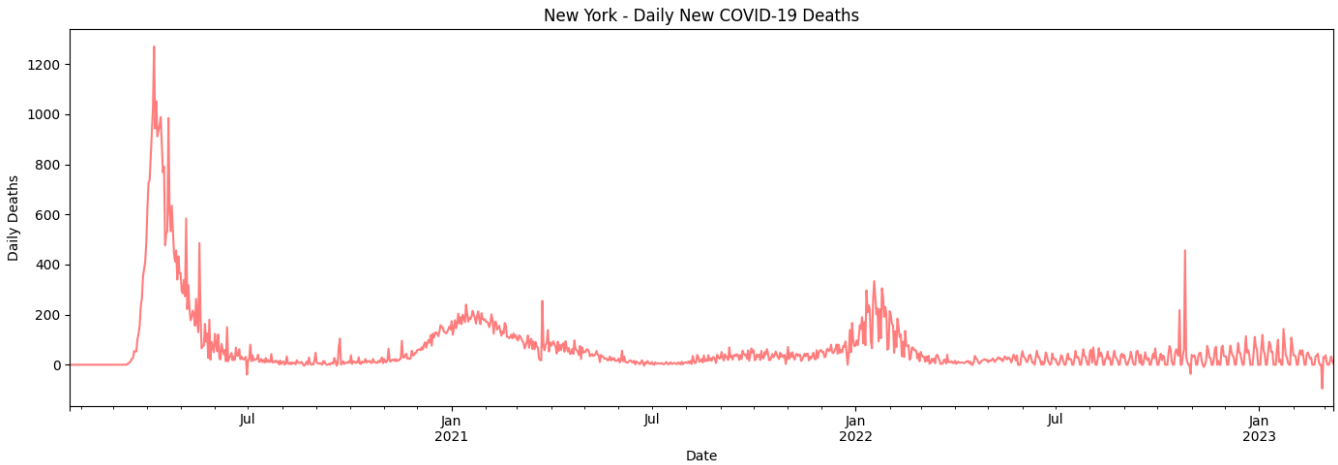

Figure 3: New York State — Daily New COVID-19 Deaths

Interpretation: New York was the epicentre of the first wave, peaking at over 1,200 daily deaths in Spring 2020. The healthcare system was overwhelmed with no prior immunity or vaccines. Subsequent waves were significantly smaller, reflecting growing immunity and vaccination. Daily deaths approached zero by 2023.

3.4 Weekly Rates of Change

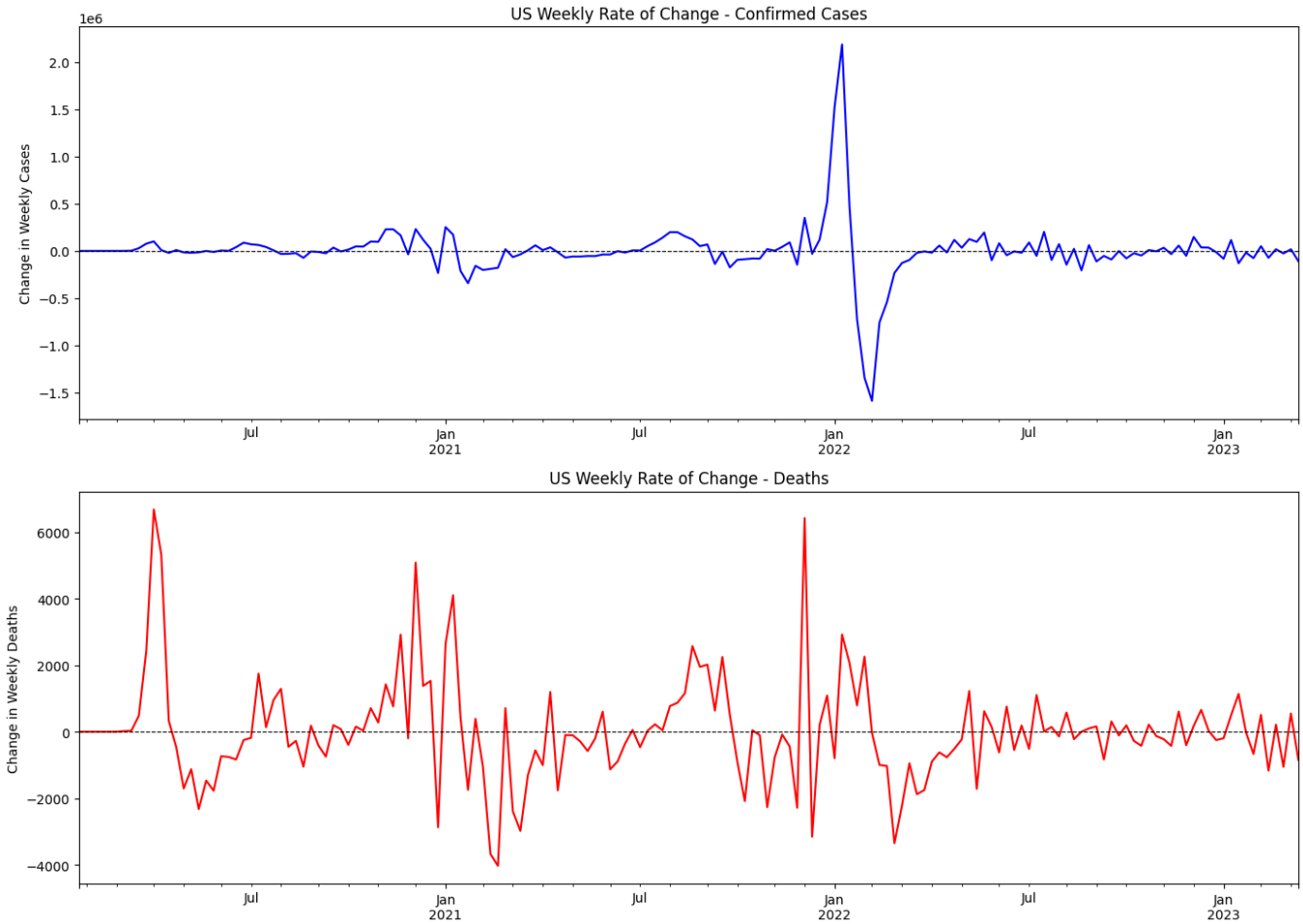

Figure 4: Weekly Rate of Change — Confirmed Cases (blue) and Deaths (red)

Interpretation: The Omicron variant produced the largest weekly acceleration, a single-week increase of over 2 million cases in early January 2022, followed immediately by a sharp drop. For deaths, the largest weekly increases occurred during the first wave and the Winter 2020–21 surge. By 2023, weekly changes will fluctuate close to zero.

3.5 Monthly Rates of Change

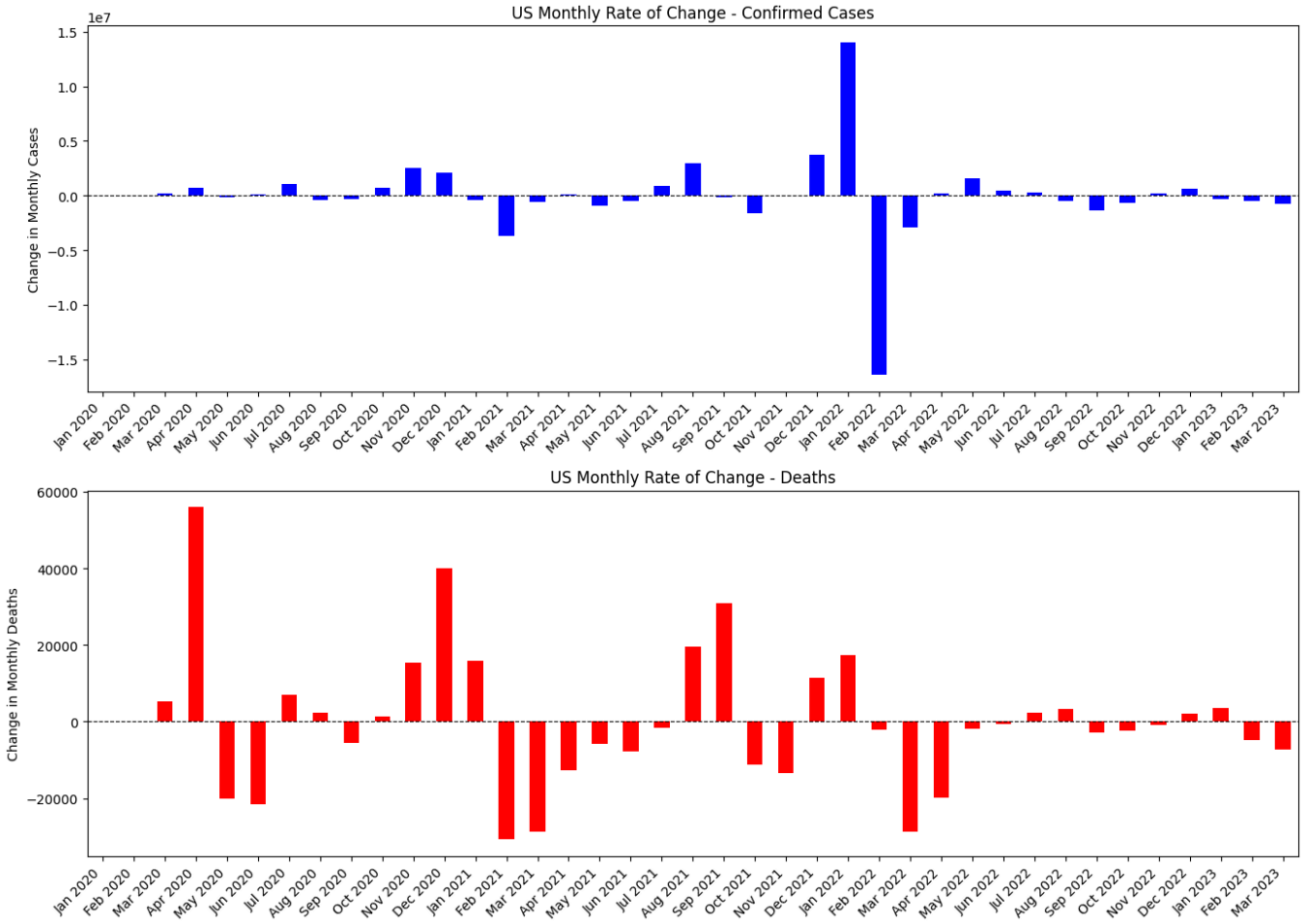

Figure 5: Monthly Rate of Change — Confirmed Cases (blue) and Deaths (red)

Interpretation: January 2022 saw the largest monthly increase in confirmed cases (+14 million), driven by Omicron. For deaths, April 2020 and December 2020 to January 2021 represent the periods of fastest escalation, each adding over 40,000 additional monthly deaths. By 2022–2023, monthly changes approach zero.

5.5 Quarterly Heatmap — Top 10 States

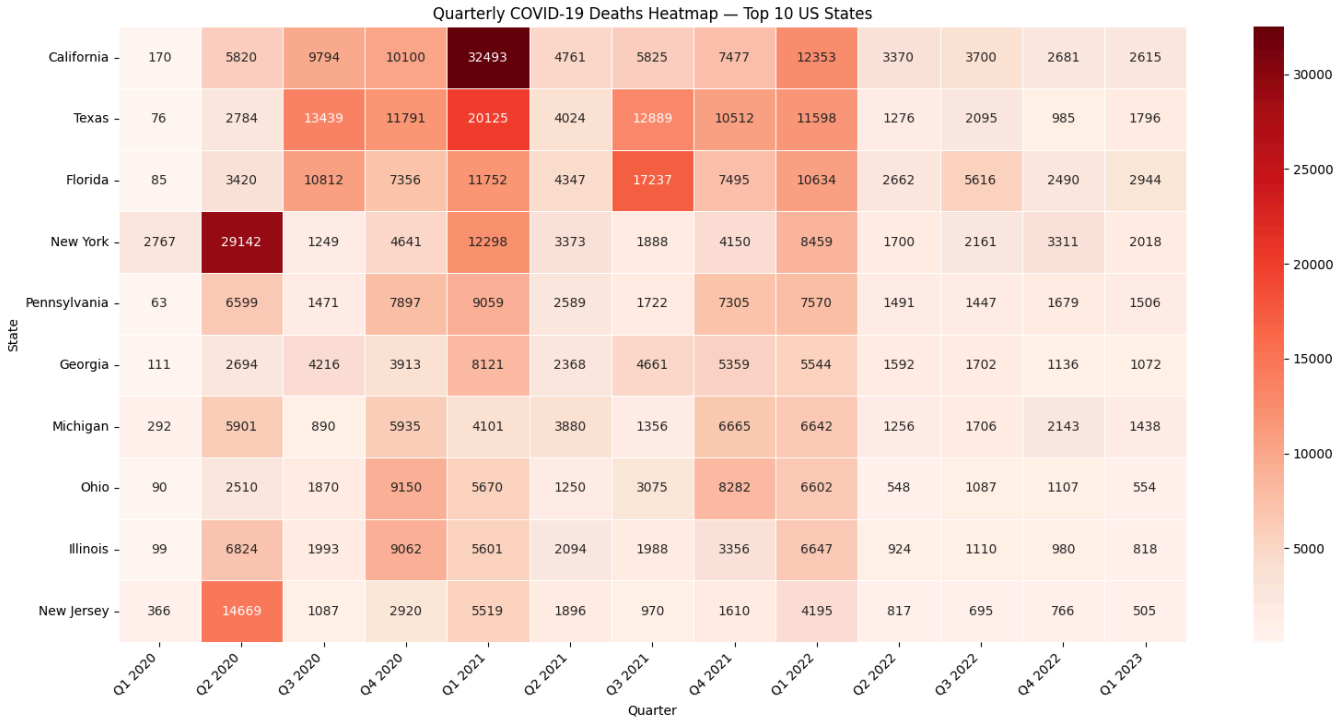

Figure 6: Quarterly COVID-19 Deaths Heatmap — Top 10 US States

Interpretation: New York recorded its worst quarter in Q2 2020 (29,142 deaths), confirming it as the first epicenter. California peaked later in Q1 2021 (32,493 deaths), suggesting a delayed first wave on the west coast. Florida and Texas peaked during the Delta variant period (Q2–Q3 2021). All states show significantly lower death counts from 2022 onwards.

4. Time Series Analysis

4.1 Rolling Mean & Standard Deviation

Before building a forecasting model, it is important to understand the distribution and volatility of the data. Rolling averages smooth out day-to-day noise to reveal the underlying trend, while rolling standard deviation measures how unpredictable the data was at any given point in time.

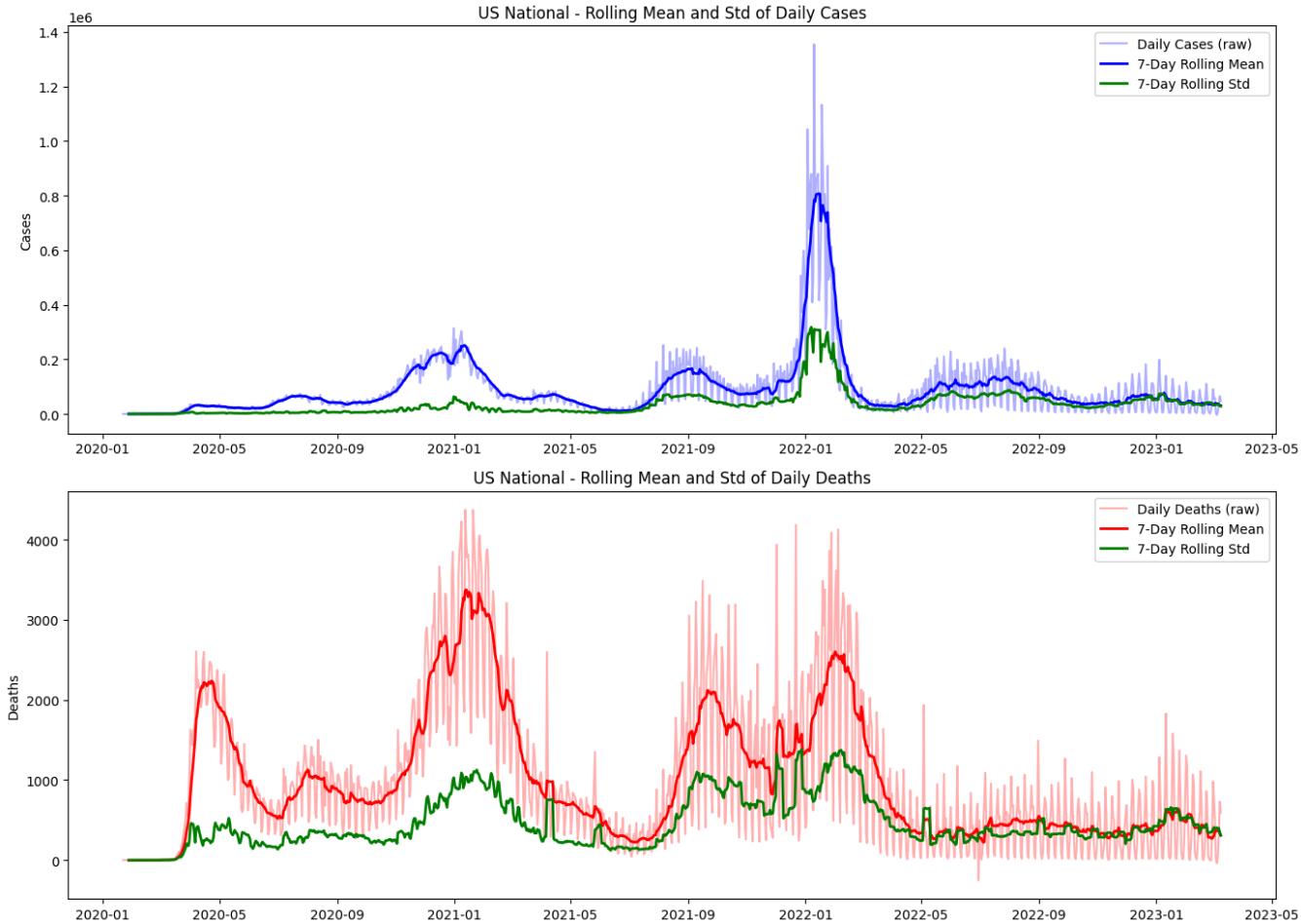

Figure 7: 7-Day Rolling Mean and Standard Deviation — US National Daily Cases (top) and Deaths (bottom)

Cases (top chart): The Omicron wave dominates this chart almost entirely, with the rolling mean peaking at over 800,000 daily cases in January 2022. All previous waves appear relatively minor in comparison. The rolling standard deviation peaks sharply during the same period, confirming this was the most volatile and unpredictable phase of the pandemic in terms of case counts.

Deaths (bottom chart): Unlike cases, deaths show a more even distribution of waves. The deadliest sustained period was Winter 2020–21, with the rolling mean reaching approximately 3,300 daily deaths. Subsequent waves produced lower peaks. By 2023, both the rolling mean and standard deviation approach near zero, a clear signal that the pandemic was ending.

4.2 Stationarity Check — Dickey-Fuller Test

The ARIMA model requires data that is stationary, meaning it fluctuates around a stable average rather than continuously trending upward or downward. Cumulative death data is clearly non-stationary (it only goes up), so we first converted it to daily new deaths using differencing.

Series	P-Value	Result
National Daily Deaths	0.0223	**Stationary**
National Daily Cases	0.0004	**Stationary**

Both series passed the test with p-values well below the 0.05 threshold. This confirms that converting from cumulative to daily data was sufficient to stabilise the series, and that the data was ready for ARIMA modelling.

4.3 ACF & PACF Analysis

The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were used to determine the appropriate parameters for the ARIMA model. These plots show how strongly today's value is related to values from previous days.

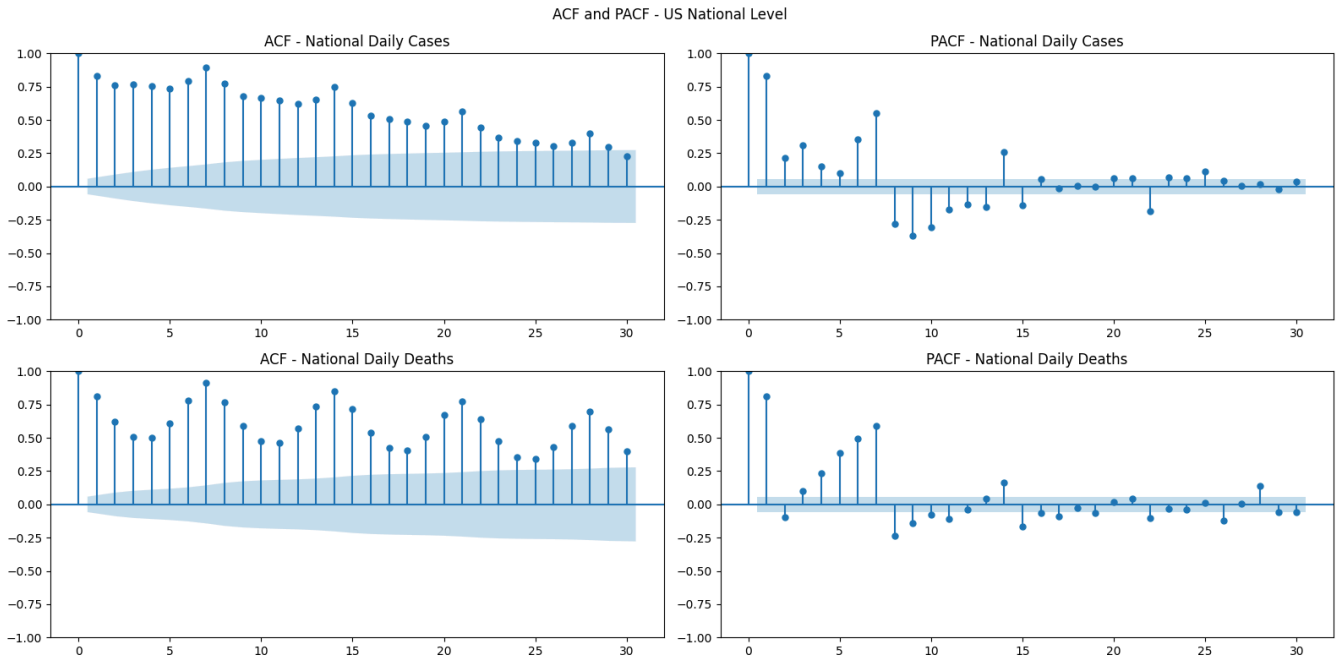

Figure 8: ACF and PACF Plots — National Daily Cases and Deaths

Both the cases and deaths plots reveal a consistent pattern of significant correlations at every 7th lag, a clear weekly cycle driven by the consistent drop in reporting over weekends. The PACF plots cutting off after the first few significant lags guided the selection of p=5 for the ARIMA model.

5. ARIMA Forecasting Model

5.1 What is ARIMA?

ARIMA stands for AutoRegressive Integrated Moving Average. It is one of the most widely used statistical models for time series forecasting. The model learns patterns from historical data and uses them to predict future values.

Parameter	Value	Meaning
p = 5	AutoRegressive	The model uses the last 5 days of data to predict the next day
d = 1	Integrated	The data was differenced once to remove the upward trend
q = 2	Moving Average	The model uses the last 2 forecast errors to correct itself

5.2 Train-Test Split

The national daily death data (1,143 days) was divided into a training set and a test set to simulate real-world forecasting conditions. The model learned from historical data and was then asked to forecast a period it had never seen.

The training period was 914 days (Jan 2020 - July 2022) and the test period was 229 days(July 2022 - March 2023.

5.3 Forecast Results

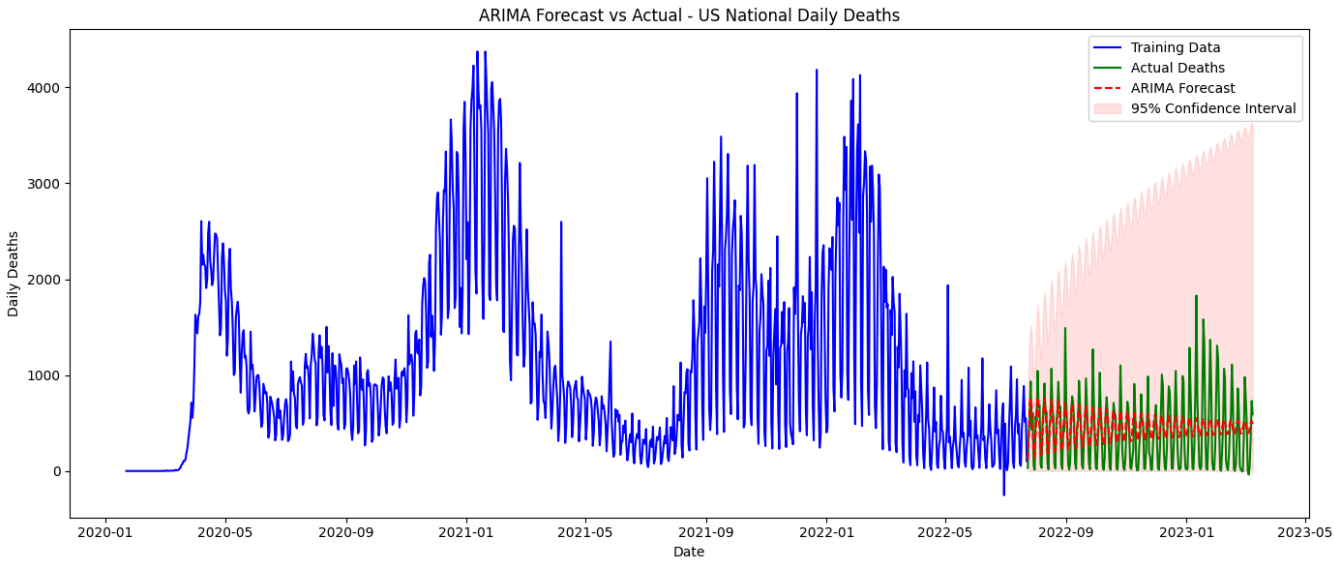

Figure 9: ARIMA(5,1,2) Forecast vs Actual — US National Daily Deaths with 95% Confidence Interval

Interpretation: The forecast (red dashed line) tracks closely with the actual deaths (green) during the test period, successfully capturing the general downward trend and the stable range of 300–600 daily deaths observed through late 2022 and early 2023.

The shaded red area represents the 95% confidence interval, the range within which the model expects the true value to fall. The interval widens progressively over time, which is expected behaviour for any forecasting model: the further into the future you predict, the less certain the prediction becomes.

Evaluation Metric	Result
MAPE	42.65%
AIC Score	13,432
Confidence Interval	95%

MAPE (42.65%): The model's predictions were off by approximately 42% on average. While this may appear high, it is an expected result for pandemic data, sudden events like new variants, policy changes, and inconsistent reporting cannot be anticipated by any mathematical model. The model correctly captured the general downward trend of deaths through the test period.

Confidence Interval: The shaded region on the forecast chart represents the 95% confidence interval. It widens progressively over time, reflecting increasing uncertainty the further into the future the model predicts. This is expected behaviour for any time series forecasting model.

6. Conclusions & Recommendations

6.1 Conclusion

Exploratory data analysis revealed that population density is the strongest geographic predictor of pandemic severity, all top 10 states by total deaths are among the most populous in the US. Time-based analysis identified the Winter 2020–21 surge and the Omicron wave (January 2022) as the two most critical periods of the pandemic. The divergence between cases and deaths during the Omicron wave provides clear evidence of vaccine effectiveness. The ARIMA model successfully forecast the downward trend of national daily deaths through early 2023, achieving a MAPE of 42.65%.

6.2 Recommendations

The model can be improved by incorporating external variables such as vaccination rates, mobility data, and policy change indicators. More advanced models such as LSTM neural networks may better capture the complex, non-linear patterns in pandemic data. A real-time monitoring dashboard that tracks weekly acceleration rates could provide early warning of emerging waves. The model should support decision-making and resource allocation, not replace public health judgment.